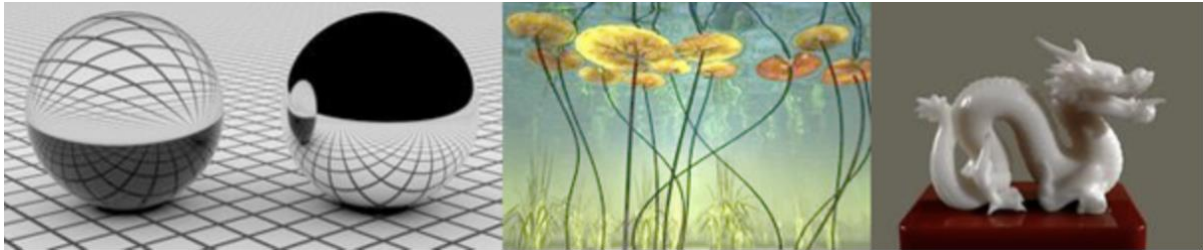


Research Project-2: Photo-Realistic Rendering Competition

(2026-S1: V7)



Purpose:

The best way to learn Computer Graphics is through developing a working 3D graphics rendering system to create stunning photo-realistic images. This is the objective of this final capstone project.

The goal of this **Rendering Competition** is to implement a rendering program in C/C++ (or Java) to render a 3D scene constructed by yourself, mimicking a given motivational image or scene.

This project is to be completed in team. Your project team will work on a moderate sized project that incorporates one or more technical ideas in computer graphics.

Each project group will choose a motivational image/scene, from the following “sample motivational images”, to create a virtual 3D scene, and render it photo-realistically. You must implement the necessary graphics features to achieve the artistic vision of the motivational image.

Two exemplar so-called “**motivational images**” are given as follows:



Your final project report must clearly summarise the **motivation, final goal, problem analysis, main methods or key graphics techniques, rendition results, and conclusions**. The report should be **self-contained**, written in the style of a short graphics conference paper.

Each project group will also deliver a **live seminar presentation** lasting up to **4 minutes**, including Q&A if any. It is recommended that you keep your main presentation within **3.5 minutes** to allow time for 1~2 questions if any. **Time-Management** will be part of the marking criteria.

DEMO VIDEO:

Before starting this project, you must watch the following Youtube Video to get an idea about what other computer-graphics students in the world (from ETH-Zurich) have created for their Computer Graphics Course's final rendering competitions:

<https://www.youtube.com/watch?v=U7m0bvXWmls>

Project-2 Rules:

1. Motivational Image and 3D Scene Creation

Choose one image from the “sample motivational images” provided below.

Your task is to create **3D models of the scene by C++ codes**, save the scene as an **OBJ or GLB** file, then load it into your **C++ program**. Render your scene at an image resolution of **at least 800x600 pixels** using any **photo-realistic rendering technique**.

2. Rendering Technique

You must implement the rendering procedure by your own C/C++ code, rather than using any off-the-shelf renderer (such as Blender, MeshLab, or any Game Engine).

The quality of your rendering code will be assessed. There are **no restrictions** on **which rendering technique** you use. For example, you may use one from the following list:

- Distribution ray tracing
- Rasterization with advanced features (e.g., shadows, environment mapping)
- Path tracing or global illumination methods.
- Gaussian Splatting or NeRF, starting from 3D mesh representation of a scene.

You may even combine **Rasterization**, with **Ray tracing**, **Monte Carlo Path Tracing**, or other **photo-realistic rendering** approaches. You may use AI based rendering techniques, provided that the starting point of your project is to construct a 3D scene using 3D meshes.

Tutors will evaluate your final project result **holistically**, based on its overall technical and visual quality. To achieve photorealism, it is strongly recommended that you use Ray Tracing, or Monte Carlo Path Tracing, rather than rasterization. In previous years, most winning teams had used the Monte Carlo Path Tracing technique for its stunning photo-realism.

Your rendered results **must include soft shadows** (e.g., using Shadow-Mapping if you used Rasterization, or other more advanced shadow-rendering techniques).

Real-time rendering time is **not required**, but your program should complete rendering **within a reasonable amount of time** — for example, **no longer than two~eight hours per frame** on a modern laptop (with, or without, multi-core or GPU acceleration).

3. User Controls and Features:

Your program must support **user control** or **parameterized input**, including:

- (i) Camera position and orientation
- (ii) Object position and pose
- (iii) Object surface materials or texture maps
- (iv) Toggle for enabling/disabling shadows
- (v) Light source position, size, and number of light sources.
- (vi) Environment map selection or other relevant options

In your **project report and seminar presentation**, explain each implemented graphics feature and include your **main rendering results**.

Requirement: To ensure you have created a 3D scene (rather than just a flat image), in your lab report (as well as in your seminar presentation), you must render the same scene from at least two different view-angles or viewpoints, and include the two (or more) results side by side in your project report. This is a must—failing to include them will lead to suitable mark reduction.

4. Programming Language and Libraries:

C or C++ or Python with PyTorch is the primary program language to be used for this project. You may use basic **OpenGL** libraries for basic drawing and I/O operations.

For the core rendering task, you must implement your own renderer. You **must not use** or call APIs of **MeshLab, Blender, Unity-3D, Godot, Unreal**, or any **other game engine** for the core rendering task. In the development phase you may use

these off-the-shelf renderers to get some inspirational ideas, or to validate the correctness of your created 3D scene and 3D models.

Sample Motivational Images:

The links below showcase example graphics rendering competitions from similar courses at other universities worldwide.

Your project's motivating scene must be selected from these examples; **In other words, motivational scenes from other sources are not permitted.**

In both your project proposal and final report, you must clearly identify your chosen reference image and provide an appropriate citation.

(1) Dartmouth College: (2016-2019)

<https://www.cs.dartmouth.edu/~rendering-competition/fa2019/>

<https://www.cs.dartmouth.edu/~rendering-competition/sp2018/>

<https://www.cs.dartmouth.edu/~rendering-competition/sp2017/>

<https://www.cs.dartmouth.edu/~rendering-competition/sp2016/>

(2) Stanford University (2005—2012)

- [Spring, 2008](#)
- [Spring, 2009](#)
- [Spring, 2010](#)
- [Spring, 2011](#)
- [Spring, 2012](#)

(3) EPFL (2017-2019)

<https://rgl.epfl.ch/pages/courses/cs440/sp17/competition2017>

<https://rgl.epfl.ch/pages/courses/cs440/sp18/competition2018>

<https://rgl.epfl.ch/pages/courses/cs440/sp19/competition2019>

(4) ETH Zurich (2014-2022)

- <https://github.com/ccetaw/ETHZ-Computer-Graphics-2022-Rendering-Competition>
- <https://cgl.ethz.ch/teaching/cg17/competition/competition.php>
- <https://cgl.ethz.ch/teaching/imsynth15/competition/competition.php>
- <https://cgl.ethz.ch/teaching/imsynth14/competition/competition.php>

Where or how to obtain or build 3D assets for your project?

First approach: To obtain 3D assets for your project, you may download free 3D mesh models from internet, e.g. **Free3D.com**, or **TurboSquid.com**, or other free 3D models that you can find online. You may also ask ChatGPT or DeepSeek or Claude.AI to give you some recommended free 3D resources on the internet.

Second approach: you may create your own stunning 3D models by using 3D authoring tools such as 3D Max, Maya, Blender etc.

Third approach: is to use 3D-generation AI tools to build 3D models for you. For example, you may use **Tencent's Hunyuan-3D (HY-3D)**, or **Microsoft's Trllis-3D**, or **Meshy-3D**.

<https://huggingface.co/spaces/tencent/Hunyuan3D-2>

<https://huggingface.co/spaces/JeffreyXiang/TRELLIS>

Or: <https://hyper3d.ai>

By the way, my former PhD students and project team have made major contributions to the first two 3D generation AI tools mentioned above. Some of my ex-staff members at Tencent Games later joined the Tencent Hunyuan-3D Project. (For context, I was formerly a Distinguished Scientist at Tencent, an AGM and the Chief Scientist for Tencent's XR Program leading a team of over 100 researchers. Tencent is the largest video game company in the world, and the owner of WeChat and QQ).

The project leader for Microsoft Trellis-3D and Microsoft Trellis-2—Jiaolong Yang (Principal Research Lead at Microsoft Research Asia)—was my former PhD student at Australian National University.

To better understand how these AI-based 3D generation systems work, you are encouraged to read our team's research paper: <https://arxiv.org/abs/2502.14247>

Project Deliverables

1. **Final Project-2 Report (PDF)**
 - Length: 5–9 pages (single-column, including references and results)
 - Minimum: 5 pages; Maximum: 9 pages
 - Must provide an **Originality Declaration** at the front page (though this page does *not* count toward the page limit).
2. **Seminar Presentation**
 - Maximum of 10 PowerPoint slides
 - Should include result images or demo videos
 - Designed for a 4-minute presentation.
3. **Source Code**
 - Submit as a ZIP file

Project Report Requirements

Your final Project-2 report should read like a **mini conference paper** and include:

- **Title and Authors**
List all team members and mark your own name with an asterisk (*)
- **Project Goal and Scope**
Clearly explain the objective of your project
- **Accomplishments**
Describe what your team implemented and achieved
- **Technical Details**
Include important implementation aspects (e.g., mip-map textures, mapping mouse clicks to 3D, relevant mathematical formulations, etc.)
- **Rendering Results and Experiments**
Present results along with meaningful comparisons or evaluations
 - You must include **two renderings of the same scene** from **different camera viewpoints or angles**
- **Discussion**
Reflect on limitations, possible improvements, and lessons learned
- **References** (if any)

Marking Criteria

Projects will be assessed holistically based on:

1. Quality of the written report
2. Visual quality of rendering results
3. Technical innovation and effort
4. Quality of the seminar presentation

To ensure fairness, at least **three independent markers** will evaluate each project, and their scores will be averaged to reduce subjective bias.

Regarding Confidential Peer Review:

Each student's final mark will be **adjusted based on peer review weighting**. The peer review **will be** submitted individually as a separate PDF file. We will announce the link for submitting your individual confidential peer review to Canvas in due time.

For now, you only need to record your team-work, and who contributed what and how much at which stage or on which tasks. This is a team project, we encourage collaboration and team work. In your future career you are almost surely will work in groups with your colleagues.

FAQs about COMP4610/8610 Final Research Project-2 (2026)

1. Can COMP4610 and COMP8610 students form a team together for the final project?

Yes. Mixed teams of COMP4610 and COMP8610 students are allowed. The only restriction is team size, which must be between 2 and 4 members.

2. Who is responsible for uploading the project proposal, final project report (PDF), and project presentation (PPT) to Wattle?

Each team member must individually submit all required documents (including their confidential peer-review scores) to Canvas individually. Although submissions from the same team may overlap, every student must upload their own copy.

This is important because it ensures that Canvas records your submission for grading, allows for individual and confidential submissions such as the peer review form, and promotes accountability within the team.

3. Who presents the final seminar on behalf of the team?

You may either nominate one team member to present your project talk, or have multiple members co-present. Tutors will assess the presentation based on its content and clarity as a whole, not based on each of your team members' presentation.

4. How will the final project be assessed?

The main assessment factors include the overall quality of your report and presentation, the innovation and technical depth of your work, and the quality of your submission.

Tutors will assess your project as if reviewing a short conference paper and attending a seminar presentation. Each project will be marked by at least three tutors, and your final grade will be the average of their scores to ensure fairness and consistency.

5. Will there be a detailed marking rubric?

Because this is a research-oriented project, the tasks are rather open-ended, and some level of subjectivity in marking is expected—just like in real research environments; we will minimise any bias by using averaged marks. This approach reflects the common practice of scientific peer review in computer science conferences and helps prepare you for research or industry work.

6. What if a team member is not contributing fairly?

Discuss the issue openly and respectfully within your team first. If the problem continues, contact

your tutors early—do not wait until the end of the project. Tutors can help mediate or offer guidance. Each student must also submit a confidential peer review form. This allows tutors to assess individual contributions fairly and adjust marks if necessary.

7. How long should the report and presentation be?

The project report must be between 5 and 9 pages, including references, figures, and citations. Keep it concise and focused.
The presentation should be 4 minutes in total.

8. Can we use AI tools like ChatGPT or GitHub Copilot?

AI tools may be used for brainstorming, idea generation, and debugging only. Do not use AI tools to code your report or to generate uncredited code. Always declare the use of AI tools clearly in your report and source code. You must include the originality declaration in your project report.

9. Are we allowed to use tools like Unity, OpenGL, or Blender?

You may not use high-level rendering engines such as Unity, Unreal, or Blender Cycles, or their APIs or libraries, to perform the rendering work. In your report, you must explain your own rendering algorithm or techniques.

The C++ framework provided in our past Computer Labs already includes a sufficient code base for many rendering tasks, including rasterization, ray tracing, and path tracing. You may call some basic OpenGL libraires for basic drawing and I/O operations.

10. Will the graphics rendering results be ranked for marking?

Your seminar presentation and rendering results will be evaluated, and relatively ranked, holistically by three teaching staff members together.

11. Can we add more objects or details to the rendered scene, and how closely should I follow the motivational image that I have selected ?

The motivational image is meant to inspire your creation. As long as your final scene is visually similar to the chosen image, your submission is valid—the contents do not need to be identical. What matters most is demonstrating your understanding and implementation of key graphics techniques learned in this course, such as texture mapping, shadow generation, ray tracing, transparency, fog or smoke rendering, and reflective materials.

The technical quality and depth of your work—as presented in your report and seminar—are more important than the appearance of the final image. A visually appealing rendering without solid technical work will not score highly.

12. Academic Integrity and Citations

Clearly identify which parts of your submission are **your own work** and which parts are **borrowed**. Cite all external sources, including code, frameworks, infrastructure, or AI tools (e.g., ChatGPT, DeepSeek). Include proper **citations** in both your **source code** and your **project report**, covering any external libraries or resources used.

Appendix: Format of “confidential peer review”:

My name is Alice, a COMP4610 student for 2026S1. Our project (titled:) has been completed by a 3-member group consisting of Alice, Ben and Charlie. From my best knowledge and assessment, I certify that the relative contributions among the team for Project-1 and Project-2 are the following, respectively

For Project-1:

Name and ID ;	Main duties and contributions for Project-1:	contribution weighting for Project-1: (add up to 100%)
Alice: u 1000023	Discussion, main coder and report writing,	40%
Ben: u 200000	Code reviewer, second coder, report writing. Focusing on the debug side.	35 %
Charlie: u1234567	Experiments, video demo preparation. Conclusion writing, Etc.	25%.

For Project-2:

Name and ID ;	Main duties and contributions for Project-2:	contribution weighting for Project-2:
Alice: u 1000023	Discussion, main coder and report writing,	30%
Ben: u 200000	Code reviewer, second coder, report writing. Focusing on the debug side.	50 %
Charlie: u1234567	Experiments, video demo preparation. Conclusion writing, Etc.	20%.

All peer reviews will be kept strictly confidential, and only known to the teaching team. The peer results raw records will be destroyed in early 2026 S2.

This is a mini- research-oriented project, not a homework assignment, so be creative and enjoy the process.

<END of Instructions for research project-2 >